Chapter 18, Problem 11PP

Problem

If a 2-MHz carrier is modulated by a 4-kHz intelligent signal, determine the frequencies of the three components of the AM signal that results.

Step-by-step solution

Step 1 of 3

Given carrier frequency is 2 M Hz, which is modulated by a modulating frequency of 4 k Hz.

$f_c = 2 \text{ MHz}$

$f_m = 4 \text{ kHz}$

Thus, the frequencies of the three components of the AM signal that results are lower sideband, carrier and upper sideband.

Step 2 of 3

Lower sideband:

$f_c - f_m = 2000000 - 4000$

$f_c - f_m = 1996000 \text{ Hz}$

Carrier:

$f_c = 2000000 \text{ Hz}$

Step 3 of 3

Upper sideband:

$f_c + f_m = 2000000 + 4000$

$f_c + f_m = 2004000 \text{ Hz}$

Therefore, the frequencies of the three components are

$\boxed{2{,}004{,}000 \text{ Hz}; 2{,}000{,}000 \text{ Hz}; 1{,}996{,}000 \text{ Hz}}$